

YOUR COMPLETE GUIDE TO BECOME
AN LLM ENGINEER

Learn LLM from Scratch to Production

168
DAYS

DAY-BY-DAY
ROADMAP

Build real-world LLM skills from the **ground up** — including training, fine-tuning, RAG, agent systems, and production deployment.



Neural Network Foundations

Build your own GPT from scratch



Tokenizer & Pretraining

Train your own model



LLM Apps & Agent Systems

Build useful AI applications



RAG, Fine-tuning & Production Deployment

Ship real-world products

```
def forward(x):  
    ...  
    return x
```



PHASE 1

Foundations

Day 1 – 28



PHASE 2

Tokenization & Pretraining

Day 29 – 56



PHASE 3

LLM Apps & Agent Systems

Day 57 – 84



PHASE 4

Mastery & Scale

Day 85 – 168

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PUBLICATION
SELF PUBLISHING BOOK

FREE SAMPLE PREVIEW

Learn LLM from Scratch to Production

The Complete Path to Becoming an LLM Engineer

A hands-on, practitioner-grade book — every chapter pairs theory, code, and a working project

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Companion: 168-Day LLM Engineer Roadmap

GitHub: github.com/MthreeMollik/Learn-LLM-from-Scratch-to-Production

First Edition · July 2026

FREE SAMPLE

What's Inside This Preview

This is a free sample of **Learn LLM from Scratch to Production** — it includes the complete Table of Contents (all 25 chapters + 3 appendices) and the full text of Chapter 1. The complete book is a ~190-page, hands-on curriculum where you build a GPT from scratch, train it, fine-tune it, wire it into an agent, and ship it to production — with every code block written to run, not just read.

Get the Full Book on Gumroad →

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Chapter 1 — Introduction: How This Journey Works

"You don't understand a thing until you can build it." — this is the one rule this book follows.

Why this book exists

Between 2023 and 2026, the number of people working with LLMs (Large Language Models) has grown fast — but most of them get stuck at one particular level. They can call an API, they can write a prompt, but they can't answer the questions that actually matter: what is the model doing internally, why did a fine-tune work (or not), why did an agent get stuck in a loop. As a result, they stay dependent on frameworks and tutorials, unable to debug or solve problems on their own.

This book's goal is to start from the opposite direction. Every layer — math, backpropagation, attention, tokenization, training, fine-tuning, agents, memory, production — we will **build by hand**. We won't start with a framework; we'll first build with raw building blocks to see what the thing actually is, and only then use the framework — because by that point, the framework won't feel like "magic" anymore, it'll feel like a familiar tool.

Who this book is for

This book assumes you're an **experienced software developer** — meaning you know the basics of programming, debugging, and system design. No prior machine learning experience is required — we'll build that from the ground up. What you do need:

- Comfort with Python (functions, classes, basic OOP)
- High-school-level algebra (variables, functions, graphs)
- A computer and an internet connection (no GPU needed to start — Google Colab's free tier is enough)
- Patience — because we won't take shortcuts

The structure of the book — three levels

People who work with LLMs can be split into three levels. This book covers all three, in order:

Level 1 — Model Builder. Someone who understands how a neural network learns, how a transformer works, how a language model is trained from scratch. Without this level, everything above it is just working with a "black box."

Level 2 — Model Customizer. Someone who can fine-tune an existing model for a specific task, give it new capabilities (tool use, memory), and align its behavior.

Level 3 — Product Builder. Someone who uses all this knowledge to build a real, reliable, evaluated, and secure product and put it in people's hands.

Most courses or tutorials start at Level 3 (API, prompt engineering) because it gives quick results. But this book starts at Level 1, because **work built on top never goes deep without the foundation underneath**. Someone who has built backpropagation by hand can spot a problem in a fine-tuning loss curve at a glance — someone who only followed tutorials cannot.

How to read this book

Every chapter follows the same structure:

1. **Plain-language concept** — the idea explained with an analogy or example, no math or code yet.
2. **Math/process** — as much as needed, no unnecessary formalism.
3. **Code** — a working, small, clean implementation of that concept. You should type and run this code yourself, not just read it.
4. **Exercise** — a task to modify or extend something yourself.
5. **Roadmap connection** — which days of the companion tracker this chapter matches.

One important piece of advice: even if reading the code makes you feel like you understand it, don't move on without writing it yourself. There's a huge difference between what the brain understands by reading and what it understands by writing with the hand. In Andrej Karpathy's words: *"you have to rebuild it yourself"* — this book is written in that same spirit.

An honest note

There's a lot of hype around AI right now — claims like "become an LLM expert in a few hours." This book makes no such promise. Deeply understanding LLMs, and doing real work with that knowledge, is about six months of serious, consistent daily work. This book, along with the companion 168-day tracker, is designed to help you maintain exactly that consistency.

In return, what you get is freedom from framework-dependency. When a new paper or model release comes out, you'll be able to read it and understand it, and reproduce it yourself if needed — because the foundation is something you built with your own hands.

What you need to get started

- Python 3.10+, pip
- PyTorch (`pip install torch`)
- Jupyter or Google Colab (for free GPU access)
- A GitHub account (to track and share your work)
- The companion progress tracker website (keep it open in your browser to check off daily work)

Moving forward

In the next chapter, we start with the smallest possible building block — a number, its derivative, and a graph. From here, the entire world of neural networks is built. It won't feel complicated — you'll be surprised how much is built from how little.

Roadmap connection: This chapter is preparation before starting Day 1. From Day 1 onward, move through it alongside Chapters 2 and 3.

ENJOYED THE SAMPLE?

Get the Full Book

This preview covered the Table of Contents and Chapter 1 only. The complete book takes you all the way from raw backpropagation to a production-shipped LLM system — 25 chapters, 3 appendices, ~190 pages, every concept built by hand before it's ever treated as a black box.

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